# Lecture #1: An Introduction to Statistics

## STAT 200 – Introductory Statistics

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# Introduction to Statistics

The goal of the discipline called statistics is to provide information so that informed decisions can be made.
The goal of this course is to enable you to filter through the statistics you encounter so that you can be better prepared for the decisions you make in your daily life.

# What is Statistics?

So, what is statistics?
Statistics is the science of gathering, describing, and analyzing data.
Statistics are the actual numerical descriptions of sample data.
Context is important for determining which meaning is being used.

# Branches of Statistics

There are many branches of statistics. The two main ones we will focus on in this class are:

1. **Descriptive Statistics:** This branch focuses on *displaying* statistics in a meaningful manner. It focuses on gathering and sorting data so that one can meaningfully summarize and display it.
2. **Inferential Statistics:** This branch involves using statistics to *estimate* population parameters.

# Branches of Statistics

Note that inferential statistics depends on descriptive statistics in two ways.
First, it requires descriptive statistics to perform the estimation.
Second, it uses descriptive statistics to "tell the story of the data".

# Descriptive Statistics

This branch focuses on displaying statistics in a meaningful manner. It focuses on gathering and sorting data so that one can meaningfully summarize and display it.
In this course, the second section is dedicated to descriptive statistics. In those sections, you will learn not only how to calculate statistics and display data, you will learn which statistics and graphics can be used to tell the story of the data.

# Descriptive Statistics

Here, you will learn:

- Univariate Statistics
  - mean
  - median
  - mode
  - standard deviation
  - variance
  - interquartile range
  - coefficient of variation

# Descriptive Statistics

Here, you will learn:

- Bivariate Statistics
  - covariance
  - correlation

# Descriptive Statistics

Here, you will learn:

- Univariate Graphics
  - histogram
  - box-and-whiskers plot (boxplot)
  - pie chart
  - bar chart

# Descriptive Statistics

Here, you will learn:

- Bivariate Graphics
  - side-by-side boxplots
  - stacked bar chart
  - mosaic plot
  - side-by-side bar charts
  - scatter plot

# Inferential Statistics

This branch involves using statistics to draw conclusions about population parameters.
In this course, we will cover inferential statistics in the fourth and fifth sections. The fourth introduces the topic, and the fifth dives deeper.
Together, the two sections cover a variety of estimation procedures and tests.

# Inferential Statistics

These include the following:

- one-sample $t$-procedure
- Wilcoxon test
- Binomial test
- chi-square test
- non-parametric bootstrap
- two-sample $t$-test
- Mann-Whitney test
- proportions test
- $F$-test
- chi-square goodness-of-fit test
- analysis of variance (ANOVA)
- chi-square test of independence
- correlation test
- linear regression

# Goals of Inferential Statistics

There are many types of analyses or goals for inferential statistics:
Exploratory analysis uses data to estimate parameters.
Confirmatory analysis uses statistics to test claims about reality (hypotheses).
Note that a "proper" analysis will use both if possible. The latter determines if the claim about the parameter is reasonable. If it is not, then the former will provide a set of reasonable values for the parameter.

# Exploratory Analysis

Exploratory analysis uses data to **estimate parameters**.
This is frequently done using **confidence intervals**. These confidence intervals are sets of values that are "reasonable" for the parameter.

**Note:**

- Confidence intervals are dependent on the data.
- They provide values for the parameter that are reasonable.

# Confirmatory Analysis

Confirmatory analysis uses statistics to **test claims about reality**.
When performing confirmatory analysis, one must have something in need of confirming. This "thing" is called a **hypothesis**.
Once the hypothesis exists, data are collected and used to determine if the hypothesis is reasonable, *given this data*.

**Note:**

- Hypothesis testing requires the claim before the data are analyzed.
- It uses the data to determine if the claim is reasonable.

# Presentation of Results

Sharing the results of your analyses is important; it allows others to learn from your analysis and to build upon your findings. Thus, you will need to be able to *properly report your findings.*

There is nothing special about the presentation; it follows the typical structure of any argumentative essay, albeit tuned to the proper use of data.

# Presentation of Results

- **Introduction:** gives purpose and summarizes the report, ends with the claim.
- **Literature Review:** provides logic behind why your claim is reasonable
- **Data and Methods:** provides the logic of your data, how it was collected, its structure; the logic of your statistical method, why it is appropriate, whether its requirements are met
- **Results and Discussion:** tells what the data tell you about the claim, what this means regarding the bigger question
- **Conclusion:** summarizes your report, provides weaknesses in your analysis, ends with the results